FABER-CASTELL
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Lecture 28-29
Linear Model & Test Error

Eg.

	160°C	180°C	
A	60	78	(+18)
B	72	66	(-6)

avg. is +12.

There is no effect of temp., until you say which catalyst.

- The interaction means that the effect of 1 var. depends on the value of the other. Parallel lines lets you summarize each factor on its own, while non-parallel lines do not. And averaging does not help.
- Linear Model
  ↳ Linear in params β_j not in the inputs.
  How X columns
  Transformaⁿ ⇒ $X_j = \log x, \sqrt{x}, x^2$
  Categorical var ⇒ $X_j = 1\ (G_i = j)$
  Basis expansion ⇒ $X_1 = x, X_2 = x^2, X_3 = x^3$
  Interacⁿ ⇒ $X_3 = X_1 \cdot X_2$
- if no interacⁿ

$$Y_i = \beta_0 + \beta_1 X_i + \gamma_1 D_1 + \varepsilon_i$$

both have their own lines, same slope but different intercept. They are ||.

if interacⁿ

$$Y_i = \beta_0 + \beta_1 X_i + \gamma_1 D_1 + \underbrace{\delta_1 (X_i D_i)} + \varepsilon_i$$

diff. slope & intercept not parallel
so the value of the categorical predictor affects the associatⁿ btw. Y & X.

Eg. salary = $4.2 + 1.35\,\text{CGPA} + 2.1 \cdot 1\{\text{MTech}\} + 0.45\,\text{CGPA} \cdot 1\{\text{MTech}\}$

4.20 → intercept → Msc with CGPA=0
1.35 → slope → per CGPA for Msc
2.10 → dummy → MTech at CGPA = 0
0.45 → interact → extra rise per CGPA pt. for MTech.

- K-Fold (Cross-Validation)
split in k parts. Fit k times, each time holding out 1 part & scoring on it. Avg. the k scores. Every obs. is used for scoring exactly once, and for fitting k-1 times.
- Small k means each fit uses less data, so test error is over-estimated (bias), large k means the k training sets overlap heavily, so the k scores are correlated & avg. is noisier. k = 5/10